

Construction of a Testable Prototype:

We began the process by gathering all the necessary materials, including hinges and wood, and started constructing the prototype. First, we focused on creating the basic structure for the arm. Carefully measuring and cutting the wood, we assembled the arm's components with precision, ensuring each piece was accurately aligned and securely fastened.

Next, we crafted a sturdy frame to ensure the arm remained stable and immovable under pressure. This involved adding reinforcements and cross braces, using high-quality materials to guarantee durability. We meticulously checked each joint and connection, making sure the frame could withstand both pushing and pulling forces without budging.

To secure this frame to the cart, we used several heavy-duty clamps, strategically placing them at key points to provide maximum stability. By meticulously adjusting each clamp, we ensured a tight and secure fit, preventing any unwanted movement during use. These clamps were crucial in maintaining the integrity of the frame, allowing it to stay firmly attached to the cart.

Finally, we conducted extensive testing to ensure the design met its objective of creating an adjustable handle that wouldn't strain Mr. Kiser's back while pushing the cart. Based on our observations, we made adjustments and refinements, fine-tuning the handle's height and angle for optimal ergonomics. The prototype proved effective, offering a practical solution that combined functionality with ergonomic design. With each iteration, we moved closer to perfecting a tool to make Mr. Kiser's tasks easier and more comfortable.











